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Subset selection based fusion for biomedical information retrieval tasks

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Abstract

To improve the effectiveness and efficiency of biomedical information retrieval by proposing ranking-based methods for selecting an optimal subset of retrieval systems for data fusion, we propose three ranking-based subset selection methods SFS (Sequential Forward Search), D&P (Diversity & Performance), and P&D (Performance & Diversity). These methods were applied in combination with the Reciprocal Rank Fusion technique. Experiments were conducted on four medical datasets from TREC, using between 62 and 125 candidate retrieval systems, and selecting up to 15 for fusion. The proposed subset selection methods significantly improved retrieval performance. Fusing the selected systems using RRF yielded improvements ranging from 10% to over 60% compared to the best individual retrieval system across the datasets. They also outperform the state-of-the-art technology by a large margin. In summary, our subset selection approach offers a practical and cost-efficient solution for biomedical information retrieval, achieving substantial performance gains while reducing computational overhead.

Keywords Data fusion, Subset selection, Information retrieval, Biomedical information retrieval, Ranking-based method

Introduction

Information retrieval has been a rapidly evolving field with the development of various technologies. Numerous retrieval models have been proposed, including the Boolean model, vector space model, latent semantic indexing, probabilistic models, learning to rank models, and many others [1]. Recently, deep learning and large language models have gained popularity in the field [2, 3]. Additionally, various techniques such as named entity recognition, link analysis, relevance feedback, and query expansion are available as optional components. Consequently, a wide range of retrieval systems can be developed for any given task.

The Text REtrieval Conference (TREC)¹ is an annual community-based evaluation workshop focused on information retrieval (IR). In many TREC retrieval tasks, dozens of research groups submit their systems' results for evaluation. With such a large number

¹<https://trec.nist.gov/>



of submissions, it becomes possible to combine multiple systems to achieve improved performance—a concept known as data fusion in IR. Numerous data fusion methods have been proposed, including CombSum [5], CombMNZ [5], Borda Count [6], Reciprocal Rank Fusion (RRF) [7], and linear combination approaches [8, 9] that employ various weight training strategies. Prior research has consistently shown that data fusion can effectively enhance retrieval performance [4].

Data fusion methods can be categorized into two groups: supervised and unsupervised approaches. Unsupervised methods, such as CombSum [5], CombMNZ [5], Borda Count [6], and Reciprocal Rank Fusion (RRF) [7], treat all component retrieval systems/results equally. These methods are easy to implement as they do not require any training. In contrast, supervised methods can achieve more effective results. The linear combination approach [8, 9] is the most popular supervised method. With proper weight training [9, 10], linear models can achieve strong performance.

While the number of constituent retrieval systems positively impacts fusion performance [11, 12], the associated computational and financial costs increase as well. Therefore, fusing as many retrieval systems as possible is not always a feasible strategy. Balancing performance and cost is often a more realistic objective in many retrieval applications. The specific challenge is: for a given number n , how can we select n retrieval systems from a larger pool of N systems to achieve the best possible fusion performance? This challenge is known as the subset selection problem [13].

As previously identified, subset selection is non-trivial and classified as an NP-hard problem [14]. When the candidate pool is relatively large, exhaustive search may not be practical due to computational constraints. The methods proposed can be categorized into three categories: ranking-based, clustering-based, and optimization-based [15]. Both ranking-based and clustering-based methods are more efficient than the last category, while optimization-based methods might be able to achieve better performance through hugely increasing computational cost.

However, the subset selection problem remains largely under-investigated in the information retrieval domain. [16] is the only work that addressed this problem with a clustering-based solution. In this paper, we propose several ranking-based methods for subset selection, where candidates are chosen one by one from the pool based on specific rules. To our knowledge, this is the first time that ranking-based methods are proposed for subset selection of information retrieval systems. Our experiments with four datasets demonstrate that these proposed methods significantly improve fusion performance, either in light of best constituent retrieval system, or the method proposed in [16]. This work demonstrates the potential for significant gains through targeted research in this area.

The remainder of this paper is organized as follows: Sect. 2 reviews prior work on data fusion and biomedical information retrieval. Sect. 3 introduces the proposed methodology, detailing a group of subset selection methods applicable to data fusion in information retrieval. Sect. 4 presents the experimental settings, results, and an evaluation of our proposed approaches, along with discussions and analyses for further insights. Finally, Sect. 5 offers concluding remarks.

Related work

This section reviews prior studies relevant to data fusion, particularly those focusing on subset selection from a large number of candidates to enhance retrieval performance. Given that the datasets used in this study pertain to biomedical information retrieval applications, we also review relevant literature in this domain.

Data fusion and subset selection

Up to now, most prior work on data fusion in information retrieval focuses solely on performance. CombSum [5], CombMNZ [5], Borda count [6], Reciprocal Rank Fusion (RRF) [7], and linear combination [9] are some of the commonly used methods. In recent years, those data fusion methods have been widely used in many different types of retrieval applications including biomedical information retrieval [16–18].

Several studies have considered both efficiency and performance of the fusion techniques in some way. Weight assignment of the linear combination method is important for the effectiveness of the fused results. MAP-related (Mean Average Precision) weighting schemes require to calculate MAP values of each constituent retrieval systems. Arguing that obtaining P@10 (Precision at 10 document levels) is cheaper than obtaining MAP values due to the fact that more relevance judgment is required for the latter, [19] used P@10 instead of MAP values from a group of training queries to weight all component retrieval models for fusion. It is found that the fusion results of using P@10 weights are comparable to that of using MAP weights.

Using more or fewer documents for the fusion operation is also an important factor that determines the efficiency of the fusion operation. Rather than using all the documents, but using some top-ranked documents in those constituent retrieval results for fusion, [20] empirically investigated how it affects the fusion results. After experimenting with three groups of results submitted to TREC between 2010 and 2012, it is found that using much fewer documents (e.g., 10% of the top-ranked documents in the whole result list including 1000 documents), a reduced time cost of 1/2 to 1/10 is achievable with marginal loss on fusion performance.

Sun et al. investigated cost-effective data fusion for information retrieval by reducing reliance on extensive relevance judgments [21]. Using sampling strategies, near-optimal fusion weights are achieved with just 5% of TREC-labelled data—only 20 queries and 500 judged documents. Experiments on six TREC datasets show that high retrieval performance is attainable with minimal manual effort, making advanced fusion methods more accessible.

Ma et al. proposed FUGER, an unsupervised framework that fuses large language models (LLMs) and traditional retrieval models (TRMs) for entity search [22]. It builds a query-centric entity-category graph and uses a weighted geometric mean model to unify relevance scores. A novel termination criterion ensures efficiency. Experiments show FUGER improves both precision and recall over LLMs, TRMs, and prior fusion methods.

Sharma et al. aimed to enhance multi-image question answering by integrating sub-modular subset selection into the MIRAGE retriever framework [23]. Query-aware functions like GraphCut and Facility Location pre-select semantically relevant images, improving scalability and retrieval performance. Anchor-based queries and data

augmentation further strengthen effectiveness, particularly in scenarios with large image collections.

As mentioned before, the number of information retrieval systems involved is a major indicator of the efficiency of the data fusion method. One intuitive approach is to select a sub-group based on their performance, for example, measured by MAP. This is the most straightforward ranking-based subset selection method.

Xu et al. proposed a clustering-based method to select a sub-group of candidates for better performance [16]. First, all candidate runs are grouped using the Chameleon clustering algorithm. Then, from each cluster, the most promising candidate is selected using Sequential Forward Selection, ensuring that only one run is chosen per cluster. Experiments conducted on two TREC datasets demonstrated the effectiveness of the proposed approach.

It is evident that [16] is the most relevant to the work in this paper. The empirical study shows that the proposed methods are more effective than it by a large margin.

Biomedical information retrieval

As an valuable subject with numerous documents, biomedical information retrieval has been investigated by many. Because of the richness of synonyms and closely related terms in the biomedical domain, synonyms and semantic relatedness between terms are key research topics. A number of resources have been created and used for improving biomedical information retrieval. The Unified Medical Language System (UMLS)² is a compendium of many controlled vocabularies in the biomedical sciences. Medical Subject Headings (MeSH)³ is a comprehensive controlled vocabulary for the purpose of indexing journal articles and books in the life sciences. Both of them have been used in biomedical information retrieval [24–27]. Some other resources including Wikipedia, National Center for Biotechnology Information (NCBI) databases [28], the International Classification of Diseases (ICD) [29], the Systematized Nomenclature of Medicine-Clinical Terms (SNOMED-CT)⁴ [30], and independently constructed resources [31] have also been used for this task.

Fusion and ensemble-based approaches have been investigated by some researchers [17, 19, 32, 33]. The fusion-based approaches have been widely used in some relevant events in TREC⁵ and CLEF⁶ such as Genomics ad hoc retrieval, medical records search, precision medicine search, eHealth consumer search, and eHealth technological assisted reviews, and so on.

Recently, pre-trained and large language models have been widely applied to biomedical information retrieval tasks. Researchers have explored both general-purpose models such as BERT [34] and T5 [35], as well as domain-specific models such as SciBERT [36], BioBERT [37], ClinicalBERT [38], and MedCPT [39]. Aftab et al. [40] investigated the use of GPT-4, a well-known large language model, for biomedical information retrieval, while [39] proposed MedCPT, a contrastively pre-trained transformer designed for zero-shot semantic retrieval in the biomedical domain. A RAG (Retrieval Augmented

²<https://www.nlm.nih.gov/research/umls>.

³<https://www.ncbi.nlm.nih.gov/mesh>.

⁴<http://www.snomed.org/snomed-ct/>

⁵Text REtrieval Conference: <https://trec.nist.gov/>

⁶Cross Language Evaluation Forum: <http://clef.isti.cnr.it/>,

Generation) based approach was investigated in [41] and [42], while both LLM and knowledge graphs were used in [43].

Tamine et al. [44] is a recent comprehensive survey on medical information retrieval.

Methodology

In this section, we present several subset selection methods that fall under the category of ranking-based methods.

A common characteristic of ranking-based subset selection methods is that candidates are chosen sequentially until a specified end condition (e.g., a predetermined number of selections) is met. Now the question is: given a total number of N results (or referred to as runs in TREC), we aim to select n of them, where $n < N$.

Assume we have a collection D of documents and a set of queries Q , with a group of retrieval results r_i (where $1 \leq i \leq N$), each produced by a different retrieval system. Each r_i provides a ranked list of documents with relevance scores for each query. We use $s(r_i, d, q)$ to denote the relevance score that document d obtains in run r_i for query q .

For an optimal combination of runs, we need to consider both performance and diversity simultaneously. Metrics such as MAP, RP, F-measure, or other commonly used metrics can be used to evaluate retrieval performance. To evaluate the diversity of a group of retrieval results, various methods can be employed. Here, based on the geometric framework [12], we use the Euclidean distance as a metric for dissimilarity of two retrieval results. The Euclidean distance between two runs r_i and r_j is defined as :

$$dis(r_i, r_j) = \sqrt{\sum_{q \in Q} \sum_{d \in D} (s(r_i, d, q) - s(r_j, d, q))^2} \quad (1)$$

Furthermore, the diversity of a group of retrieval results can be measured by dissimilarity of all possible pairs in that group. According to [12], both dissimilarity of two retrieval results and performance of a retrieval result can be represented by the same metric-Euclidean distance. Thus data fusion methods such as averaging (same as Comb-Sum) and linear combination can be formally defined in the geometric framework and many interesting properties can be deduced.

The average distance between a run r and a set of runs R is defined as:

$$dis(r, R) = \frac{1}{|R|} \sum_{r_j \in R} dis(r, r_j) \quad (2)$$

Equation 2 is the key point for the two subset selection methods D&P, and P&D. When we try to add a new run r into a set of runs R , $dis(r, R)$ is a useful metric for us to consider the diversity of the whole group. We will discuss that later in the paper.

Sequential Forward Search (SFS) is the first proposed method. Initially, we selected the one with the best performance in a predefined metric such as MAP. Next we fuse it with each of the remaining runs and choose the one that can lead to the best fusion performance. This process is repeated until we have a sufficient number of constituent runs.

Although diversity is not explicitly considered in the above process, it is implicitly accounted for because both performance and diversity impact fusion performance when each run is added for fusion. Refer to Algorithm 1 for more detailed description.

Input: R_{in}, n

Output: R_{out}

1. $R_{out} = \{\}$;
2. Evaluate all the runs in R_{in} to calculate their MAP values;
3. Select the best run in R_{in} based on MAP and move it to R_{out} ;
4. For each run in R_{in} , fuse it with all the runs in R_{out} ;
5. Select the run in R_{in} with the best fusion performance in MAP;
6. Move the run selected in Step 5 from R_{in} to R_{out} ;
7. Repeat Steps 4-6 until there are n runs in R_{out} .

Algorithm 1 Sequential Forward Search

Input: R_{in}, n, g

Output: R_{out}

1. $R_{out} = \{\}$;
2. Evaluate all the runs in R_{in} to calculate their MAP values;
3. Select the best one in R_{in} in MAP and move it to R_{out} ;
4. For each run r_i in R_{in} , calculate $dis(r_i, R_{out})$ by Eq. 2;
5. Set T to be g runs r_j in R_{in} with the largest $dis(r_j, R_{out})$ values;
6. For each run in T , fuse it with all the runs in R_{out} ;
7. Select one run in T with the best fusion performance in MAP;
8. Move the selected in Step 7 from R_{in} to R_{out} ;
9. Repeat Steps 4-8 until there are n runs in R_{out} .

Algorithm 2 Diversity & Performance

Input: R_{in}, n, θ

Output: R_{out}

1. Initialise $R_{out} = \{\}$;
2. Evaluate all runs in R_{in} to calculate their MAP values;
3. Select the run in R_{in} with the highest MAP and move it to R_{out} ;
4. For each run in R_{in} , fuse it with all the runs in R_{out} ;
5. Calculate each fusion performance in MAP in Step 4 and find the best p_b ;
6. If R_{out} has $n-1$ element, add the run leading to best fusion performance, stop;
7. Select all the runs in R_{in} into T whose fusion performance $p_f \geq \theta * p_b$;
8. From T , select the run r_i with the largest $dis(r_i, R_{out})$ value;
9. Move the run selected in Step 8 from R_{in} to R_{out} ;

Algorithm 3 Performance & Diversity

Diversity & Performance (D&P) is the second proposed method. In each iteration, a single run is selected. As shown in Algorithm 2, the selection process consists of two stages. First, a small group of g candidate runs is selected based on their dissimilarity to the already selected runs (Steps 4 and 5). Then, from this candidate group T , the run with all those selected that offers the best performance is chosen (Steps 6–8). In particular, the function

Performance & Diversity (P&D) is the third method. Similar to Algorithm 2, in each iteration, Algorithm 3 selects one run. This process is also divided into two stages. First, we select a small group of runs, each of which leads to the best or near-best fusion performance (Steps 4–6). Then, from this small group, we choose one run that is the most dissimilar to those already selected (Steps 7 and 8).

Example 1 Now let us use a small example to illustrate how Algorithm 3 works. Assume there are 8 runs $r_1, r_2, \dots, r_8$ in R_{in} . The requirement is to select 3 runs for best possible fusion performance using Algorithm 3. θ is set to 0.95. First we evaluate all the runs in R_{in} in MAP and select the best performer r_8 and move it to R_{out} (Steps 2 and 3). Then

there are seven runs $r_1, r_2, \dots, r_7$ remaining in R_{in} . We let each of them fuse with the one in R_{out} and evaluate the fusion performance (Step 4). It is found that the pair (r_1, r_8) obtains the best fusion performance (Step 5), pairs (r_2, r_8) and (r_3, r_8) obtain over 95% of the effectiveness of (r_1, r_8) , but other pairs do not reach this threshold (Step 7). Therefore, we further consider these three pairs (r_1, r_8) , (r_2, r_8) and (r_3, r_8) from the diversity aspect. We find that the pair (r_1, r_8) has the largest dissimilarity. Therefore, r_1 is selected and it is moved from R_{in} to R_{out} (Steps 8-9). Now six runs $r_2, r_3, \dots, r_7$ remain in R_{in} . We let each of them fuse with the runs in R_{out} and evaluate the fusion performance (Step 4). We find that the triplet (r_2, r_1, r_8) leads to the best fusion performance (Step 5). This time, because there are already 2 runs in R_{out} , diversity is no longer considered any more. Therefore, r_2 is selected and moved from R_{in} to R_{out} (Step 6). The final selection is (r_2, r_1, r_8) .

Algorithm complexity analysis

Now let us examine the complexity of these three algorithms. In Algorithm 1, with a total number of N constituent runs, we need to fuse N times with 1 run, $(N - 1)$ times with 2 runs, ..., and $(N - (n - 1))$ times with n runs. To fuse m runs, the required time is $O(m * |Q| * |D_r|)$, where $|Q|$ is the number of queries and $|D_r|$ is the number of retrieved documents in each run for a given query (assuming it is always the same). Therefore, the complexity of Algorithm 1 is:

$$\begin{aligned} C_{A1} &= O(N * 1 * |Q| * |D_r|) + O((N - 1) * 2 * |Q| * |D_r|) \\ &\quad + \dots + O((N - (n - 1)) * n * |Q| * |D_r|) \\ &= O((n^2 N - n^3) * |Q| * |D_r|) \\ &= O((n^2 N) * |Q| * |D_r|) \end{aligned} \quad (3)$$

In Algorithm 2, at each step g runs are selected by calculating the Euclidean distance using Eqs. 1 and 2. According to Eq. 1, calculating $dis(r_i, r_j)$ requires time $O(|D| * |Q|)$. Here $|D|$ is the number of documents in the whole collection. However, in practice, each retrieval system only retrieves a small number of documents. For those not retrieved documents, a score of zero is assigned by default and we do not need to consider them in Eq. 2. Therefore, $dis(r_i, r_j)$ requires time $O(|D_r| * |Q|)$ instead. $dis(r, R)$ requires $O(|R| * |D_r| * |Q|)$ time, where $|R|$ is the number of runs in R (Eq. 2). When we already have $|R|$ runs, selecting the next one requires $O((N - |R|) * |R| * |D_r| * |Q| * g) + O(|D_r| * |Q| * g)$ time. The first part is for selecting g runs by calculating the Euclidean distance, while the second part is for fusing those selected runs. Finally, the time complexity of Algorithm 2 is $C_{A2} = O((n^2 N) * |Q| * |D_r| * g)$.

The time complexity of Algorithm 3 is related to the input parameter θ . Assume that the performance of all the runs are evenly distributed. Therefore, θ interprets as a number of $g_2 = (1 - \theta) * N$ runs for the group. The complexity of Algorithm 3 is $C_{A3} = O((n^2 N) * |Q| * |D_r| * g_2)$.

Experiments

This section presents the datasets, experimental settings, and results of our empirical study. Finally, we provide a sensitivity analysis of the parameters used in Algorithms 2 and 3.

Table 1 Statistics of the four TREC datasets used in the experiment

Year	Research teams	Runs	Queries	Best MAP	Average MAP
2017	29	125	30	0.2327	0.1184
2018	24	103	50	0.3296	0.1922
2019	14	62	40	0.3303	0.2069
2020	16	66	40	0.4078	0.2668

Table 2 Technologies used in the precision medicine track of TREC 2020

Group	Technologies used (F: first round; R: re-ranking)
Aliababa	F: Elasticsearch +query expansion R: BioBERT (citation, expert annotations)
BIT	F: BM25 R: neural IR
SIB	F: Elasticsearch with two query formats R: fusion, kNN, SVC, publication type, group size, etc
CincyMedIR	F: Elasticsearch R: BM25+random forest
CSIROmed	F: DFR R: BERT-based neural models
H2oloo	F: BM25 R: T5 sequence-to-sequence transformer
IMS	F: BM25, DFR, the language model R: fusion method RRF
UWaterloo	F: BM25, pseudo-feedback, manual feedback R: not applied
uogTr	F: DFR R: SciBERT and ColBERT
VOH	F: BM25 from Elasticsearch, query expansion R: not applied

Elasticsearch <https://www.elastic.co/>

Datasets

Between 2017 and 2020, the Text REtrieval Conference (TREC) organized the precision medicine track for four consecutive years. This track typically included two sub-tasks. Our study focuses on the literature article search sub-task. The corpus consisted of 26,759,399 MEDLINE abstracts and two additional sets of abstracts: (i) 37,007 abstracts from the proceedings of the American Society of Clinical Oncology (ASCO), and (ii) 33,018 abstracts from the proceedings of the American Association for Cancer Research (AACR) [45].

Each year, a set of queries was chosen and released, allowing research groups to submit results (runs). Each research group was permitted to submit up to five runs. Typically, the runs submitted by the same research group were obtained by running the same retrieval system with different components, settings, query formats, and so on. Some related statistics of the four data sets are presented in Table 1.

It would be interesting to find the methodologies used in those participant information retrieval systems. They are good representatives of the technologies up to that point. In order to archive that, we read the reports of those groups that submitted runs to the Precision Medicine Track in TREC 2020. In total 16 research teams submitted 66 runs. Among them, 10 presented their technical reports in the proceedings of TREC. Table 2 presents related technologies used for generating these runs.

Typically, a two-stage process (first-round retrieval and re-ranking) is applied in these retrieval systems to fulfil the task. From Table 2, we can see that during the first round, all of them used traditional retrieval models, including BM25, DFR (Divergence From Randomness), and language models. Additionally, query expansion is commonly employed.

At the re-ranking stage, machine learning technologies are widely used. BERT-related technologies (such as BioBERT, SciBERT, and ColBERT), T5, other neural IR models, and fusion/ensemble methods are utilized. Some retrieval systems also incorporate special features such as citation data, expert annotations, publication type, and author-related meta-information.

This indicates that up to 2000, both traditional retrieval methods and machine learning technologies are essential components for constructing retrieval systems in the biomedical domain. The same observation is also confirmed in some more recent publications, e.g., in [34, 46].

Experimental settings

First we introduce two ranking-based methods that will be used as baseline methods in our experiments:

- T-MAP: This method ranks all the runs by their Mean Average Precision (MAP) scores and selects the top n runs based on these MAP scores.
- G-MAP: Similar to T-MAP, G-MAP ranks all the runs by their MAP scores and selects the top n runs. However, G-MAP includes an additional constraint: only one run from each participant group is allowed. This method is particularly suitable for scenarios like TREC submissions. Since runs submitted by the same participant tend to be more similar to each other than to those submitted by different participants, it makes sense to group the runs by their submitters.

These two methods are straightforward to implement yet offer reasonably good performance, serving our purpose well.

The clustering-based method C-S (Clustering-Sequential Forward Search), proposed in [16], will be served as the third baseline method.

In each dataset, we select 1, 2, ..., up to 15 runs from all the submitted runs using the selection algorithms discussed in Sect. 3. The fusion results are evaluated using four metrics: MAP, RP, P@10, and P@20.

Score normalization is an important step for data fusion methods to work properly. In this study, we use the RRF method [7] to convert rankings into scores. For a ranked list of documents related to a specific query retrieved by a retrieval system, the score of a document at ranking position i is defined as:

$$s(d, rank) = \frac{1}{60 + rank} \quad (4)$$

In this way, the document at ranking position 1 gets a score of 1/61, the document at ranking position 2 gets a score of 1/62, and so on.

For the selected runs, we applied CombSum(RRF) and CombMNZ(RRF) to fuse the results. In both methods, the Reciprocal Rank Fusion (RRF) technique is used to convert rankings into scores. CombSum(RRF) then combines these scores using the CombSum

formula, while CombMNZ(RRF) applies the CombMNZ formula for score aggregation. For simplicity, we refer to them as CombSum and CombMNZ throughout the remainder of this paper.

We evaluated the three proposed methods using all four datasets. In Algorithm 2, the parameter g is set to 5 for D&P. In Algorithm 3, the parameter θ is set to 0.995 for P&D. See Sect. 4.4 for detailed analysis about them. Two other methods T-MAP and G-MAP serve as baseline methods. Besides, C-S is used as a baseline method for the 2017 and 2018 datasets, with results copied from [16].

Experimental results

The experimental results using the CombSum method across all four metrics are presented in Table 3. For each method and metric, the table reports the average fusion performance on a given dataset, computed over all queries and using 2 to 15 component runs.

As shown in the results, the three proposed methods—SFS, D&P, and P&D—consistently outperform the two baseline methods, G-MAP and T-MAP, by a substantial margin. The proposed methods yield comparable performance to each other, while the performance gap between the proposed and baseline methods varies across different metrics. For MAP and RP, the improvement is approximately 10%, whereas for P@10 and P@20, it is around 5%. Notably, these differences also vary across datasets. Among the proposed methods, P&D and D&P achieve the best performance in eight and seven cases, respectively, while SFS outperforms the others in two cases.

We also present the fusion results for CombMNZ in Table 4. Compared with CombSum, the results for CombMNZ are generally not as good. However, in a few cases, CombMNZ obtains better results than CombSum, mainly on P@10. This is quite interesting. Additionally, similar to CombSum, the three proposed methods SFS, D&P, and P&D perform better than the two baseline methods G-MAP and T-MAP by a large margin, while the three proposed methods are quite close to each other. In the following we focus on CombSum.

Table 3 Performance of a group of subset selection algorithms (CombSum, 2017–2020, number in bold indicates the best performance)

Year	Metric	G-MAP	T-MAP	SFS	D&P	P&D
2017	MAP	0.3212	0.2863	0.3575	0.3566	0.3603
	RP	0.3669	0.3370	0.3920	0.3928	0.3947
	P@10	0.6417	0.6462	0.6960	0.6871	0.6957
	P@20	0.5644	0.5762	0.6076	0.6123	0.6156
2018	MAP	0.4071	0.3899	0.4516	0.4520	0.4527
	RP	0.4292	0.4163	0.4595	0.4600	0.4598
	P@10	0.6729	0.6947	0.7417	0.7446	0.7404
	P@20	0.6335	0.6424	0.6884	0.6902	0.6903
2019	MAP	0.3729	0.3233	0.4384	0.4384	0.4381
	RP	0.3894	0.3564	0.4392	0.4402	0.4391
	P@10	0.6188	0.6164	0.6943	0.6982	0.6964
	P@20	0.5953	0.5831	0.6497	0.6495	0.6502
2020	MAP	0.4357	0.4396	0.4912	0.4929	0.4928
	RP	0.4300	0.4513	0.4853	0.4847	0.4859
	P@10	0.5613	0.5774	0.6044	0.6071	0.6081
	P@20	0.4934	0.4985	0.5298	0.5318	0.5338

Table 4 Performance of a group of subset selection algorithms (CombMNZ, 2017–2020, number in bold indicates the best performance, number in italics indicates better performance than its CombSum counterpart)

Year	Metric	G-MAP	T-MAP	SFS	D&P	P&D
2017	MAP	0.3094	0.2774	0.3362	0.3360	0.3389
	RP	0.3552	0.3320	0.3727	0.3746	0.3778
	P@10	0.6348	<i>0.6464</i>	0.6679	0.6671	0.6776
	P@20	0.5576	0.5704	0.5802	0.5826	0.5921
2018	MAP	0.4015	0.3879	0.4366	0.4372	0.4371
	RP	0.4242	0.4140	0.4473	0.4482	0.4480
	P@10	<i>0.6757</i>	0.6939	<i>0.7420</i>	0.7427	0.7434
	P@20	0.6367	0.6424	0.6851	0.6875	0.6864
2019	MAP	0.3580	0.3196	0.4144	0.4139	0.4147
	RP	0.3792	<i>0.3568</i>	0.4194	0.4195	0.4204
	P@10	<i>0.6244</i>	<i>0.6220</i>	0.6875	0.6880	0.6884
	P@20	0.5938	0.5830	0.6396	0.6395	0.6429
2020	MAP	0.4198	0.4287	0.4685	0.4663	0.4686
	RP	0.4204	0.4426	0.4659	0.4663	0.4677
	P@10	0.5525	0.5740	0.5986	0.5899	0.5952
	P@20	0.4901	0.4969	0.5229	0.5260	0.5281

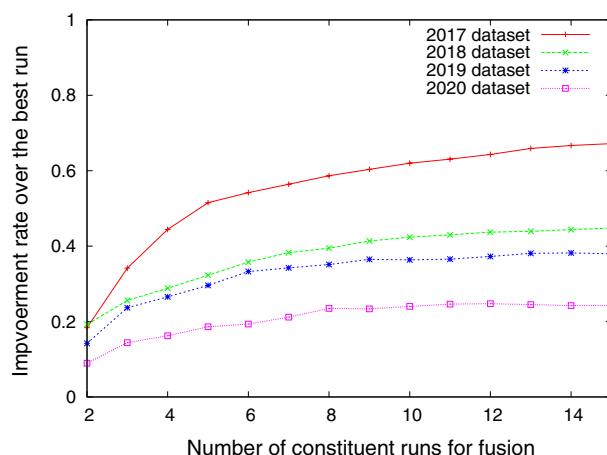


Fig. 1 Fusion performance improvement over the best run (P&D, MAP)

Next, we compare the performance of P&D (in terms of MAP) with that of the best individual component run. The results are shown in Fig. 1. It is evident that the fused results produced by P&D outperform the best component runs across all four datasets. However, the magnitude of improvement varies significantly across datasets.

One possible explanation is that as both the best and average performance of individual runs have improved over the years, achieving further gains through fusion has become more challenging. In 2017, improvement rates range from just under 20% (when fusing two runs) to over 60% (when fusing 15 runs). In contrast, in 2020, the improvement rates range from about 10% (fusing two runs) to over 20% (fusing 15 runs).

For more details about the experiments, Tables 5, 6, 7 and 8 present the performance (in MAP) of all methods across different numbers of constituent fusion runs in each dataset. SFS, D&P, and P&D outperform the other methods in all cases except one (2017, selecting two runs), where G-MAP performs as well as SFS and D&P. Comparing any of

Table 5 Performance of a group of subset selection algorithms (CombSum, MAP, 2017)

Num	G-MAP	T-MAP	C-S	SFS	D&P	P&D
2	0.2757	0.2425	0.2417	0.2757	0.2757	0.2754
3	0.2926	0.2388	0.2735	0.3061	0.3061	0.3121
4	0.3015	0.2411	0.3038	0.3361	0.3337	0.3360
5	0.3280	0.2632	0.3110	0.3498	0.3480	0.3525
6	0.3208	0.2728	0.3080	0.3549	0.3556	0.3588
7	0.3296	0.2857	0.3076	0.3607	0.3629	0.3639
8	0.3201	0.2924	0.3116	0.3646	0.3648	0.3691
9	0.3302	0.2908	0.3031	0.3686	0.3683	0.3731
10	0.3299	0.3007	0.3036	0.3717	0.3730	0.3769
11	0.3298	0.3131	0.3076	0.3758	0.3761	0.3794
12	0.3284	0.3074	0.3159	0.3805	0.3788	0.3823
13	0.3367	0.3155	0.3117	0.3845	0.3809	0.3860
14	0.3379	0.3211	0.3093	0.3866	0.3831	0.3878
15	0.3349	0.3221	0.3041	0.3882	0.3848	0.3890
Ave.	0.3212	0.2863	0.3009	0.3574	0.3566	0.3603
Imp.			+0.0%	+18.18%	+18.51%	+19.74%

Number in bold indicates the best performance

Table 6 Performance of a group of subset selection algorithms (CombSum, MAP, 2018)

Num	G-MAP	T-MAP	C-S	SFS	D&P	P&D
2	0.3886	0.3298	0.3361	0.3930	0.3930	0.3929
3	0.4085	0.3293	0.3837	0.4139	0.4139	0.4139
4	0.4029	0.3679	0.3950	0.4249	0.4249	0.4246
5	0.4003	0.3840	0.3914	0.4354	0.4346	0.4360
6	0.3948	0.3882	0.4002	0.4474	0.4421	0.4474
7	0.4120	0.3884	0.3985	0.4554	0.4554	0.4557
8	0.4244	0.3992	0.4062	0.4591	0.4590	0.4596
9	0.4179	0.4084	0.4010	0.4660	0.4659	0.4658
10	0.4177	0.4089	0.4194	0.4697	0.4697	0.4692
11	0.4126	0.4140	0.4142	0.4712	0.4712	0.4712
12	0.4093	0.4098	0.4144	0.4729	0.4729	0.4737
13	0.4068	0.4128	0.4221	0.4698	0.4737	0.4744
14	0.4008	0.4089	0.4221	0.4712	0.4752	0.4758
15	0.4017	0.4086	0.4230	0.4728	0.4764	0.4772
Ave.	0.4071	0.3899	0.4020	0.4516	0.4520	0.4527
Imp.			+0.0%	+12.34%	+12.44%	+12.61%

Number in bold indicates the best performance

the three proposed methods with any of the baseline methods, the difference is significant at the level of 0.01 (paired-samples t-test).

The results of C-S are taken from [16]. Only two datasets, 2017 and 2018, were used in that study [16]. Therefore, we present the results of it in these two datasets and compare it with the others in these two datasets separately. Overall, it is better than T-MAP and slightly worse than G-MAP. It does not perform as good as the three proposed methods, which is between 18 and 20% for the 2017 dataset and between 12 and 13% for the 2018 dataset. The average improvement rate of the three proposed methods over C-S is presented in the last line in Tables 5 and 6.

Finally, we report the execution time for the subset selection methods in Table 9. Among the four dataset used for the experiment, we chose two years: 2017 and 2019. The year 2017 has the most runs (125), while 2019 has the least number of runs (62). A

Table 7 Performance of a group of subset selection algorithms (CombSum, MAP, 2019)

Num	G-MAP	T-MAP	SFS	D&P	P&D
2	0.3530	0.3303	0.3772	0.3772	0.3772
3	0.3518	0.3603	0.4084	0.4079	0.4084
4	0.3697	0.3531	0.4179	0.4175	0.4179
5	0.3680	0.3395	0.4280	0.4273	0.4280
6	0.3715	0.3304	0.4401	0.4404	0.4401
7	0.3950	0.3238	0.4433	0.4436	0.4433
8	0.3859	0.3177	0.4465	0.4469	0.4462
9	0.3793	0.3139	0.4503	0.4505	0.4508
10	0.3806	0.3104	0.4501	0.4503	0.4504
11	0.3760	0.3075	0.4514	0.4524	0.4509
12	0.3686	0.3047	0.4553	0.4553	0.4533
13	0.3736	0.3086	0.4560	0.4558	0.4560
14	0.3737	0.3107	0.4563	0.4562	0.4564
15	NA	0.3151	0.4557	0.4559	0.4557

Number in bold indicates the best performance

Table 8 Performance of a group of subset selection algorithms (CombSum, MAP, 2020)

Num	G-MAP	T-MAP	SFS	D&P	P&D
2	0.4396	0.4396	0.4443	0.4443	0.4443
3	0.4413	0.4391	0.4665	0.4665	0.4665
4	0.4556	0.4388	0.4739	0.4739	0.4739
5	0.4580	0.4406	0.4836	0.4836	0.4836
6	0.4451	0.4350	0.4876	0.4864	0.4866
7	0.4399	0.4454	0.4908	0.4921	0.4940
8	0.4418	0.4402	0.4955	0.5017	0.5034
9	0.4369	0.4350	0.4975	0.5018	0.5030
10	0.4272	0.4413	0.5012	0.5059	0.5057
11	0.4325	0.4378	0.5032	0.5069	0.5082
12	0.4238	0.4367	0.5071	0.5081	0.5087
13	0.4150	0.4406	0.5091	0.5084	0.5076
14	0.4254	0.4421	0.5085	0.5113	0.5066
15	0.4170	0.4409	0.5068	0.5094	0.5066

Number in bold indicates the best performance

Table 9 Execution time for subset selection methods (unit: ms)

Year	Runs selected	T-MAP	SFS	D&P	P&D
2017	2	6354	57,766	69,485	70,102
2017	8	6437	397,961	429,534	429,782
2017	15	6583	998,359	1,050,892	1,066,639
2019	2	4099	33,650	38,683	38,339
2019	8	4135	135,880	152,610	154,707
2019	15	4165	371,714	398,432	388,158

personal computer with i7-10700 CPU @ 2.90 GHZ and 16 G main memory was used. In all cases, both D&P and P&D take approximately the same amount of time, while SFS is slightly quicker. In the 2017 dataset, to select 15 from a total of 125 runs, it takes slightly over 1000 s for either D&P or P&D to complete. For the same task, less than 40% of the time is required for the 2019 dataset.

Parameter sensitivity analysis

One key parameter is defined in both Algorithms 2 and 3: it is g in Algorithm 2 and θ in Algorithm 3. In this section, we analyse the impact of these parameters on their respective algorithms.

In Algorithm 2, g represents the size of the group T after the first-stage selection. This group T consists of the g most diverse candidates compared to the current selection. To better understand the role of g , consider two extreme cases:

- Case 1: Set $g=1$. In this scenario, only one candidate is passed to the second stage, effectively removing performance as a selection criterion. The algorithm becomes purely diversity-driven.
- Case 2: Set g as large as possible, such that T includes all possible candidates. Here, the algorithm reduces to a purely performance-based selection method – similar to T-MAP, if MAP is used as the retrieval evaluation metric.

Therefore, it is important to investigate how varying g influences the performance of the D&P algorithm.

Figure 2 shows how data fusion performance changes with different values of g in Algorithm 2. We observe that all four performance curves rise sharply at first, peak around $g = 5$ or $g = 6$, and then decline gradually. The similarity among these curves suggests that setting g to be 5 or 6 is generally a good choice in similar scenarios.

In Algorithm 3, θ determines the size of the group T after the first-stage selection. When $\theta=1$, the algorithm reduces to a purely performance-based selection method—such as T-MAP. Conversely, when $\theta=0$, it becomes a purely diversity-based selection method. Therefore, it is also worthwhile to investigate how varying θ impacts the performance of the P&D algorithm.

Figure 3 shows how data fusion performance varies with different values of θ in Algorithm 3. All four performance curves increase gradually at first, reach their peak around $\theta = .99$ or $\theta = .995$, and then decline rapidly. The consistency among these curves suggests that setting θ to be 0.99 or 0.995 is generally an effective choice in similar scenarios.

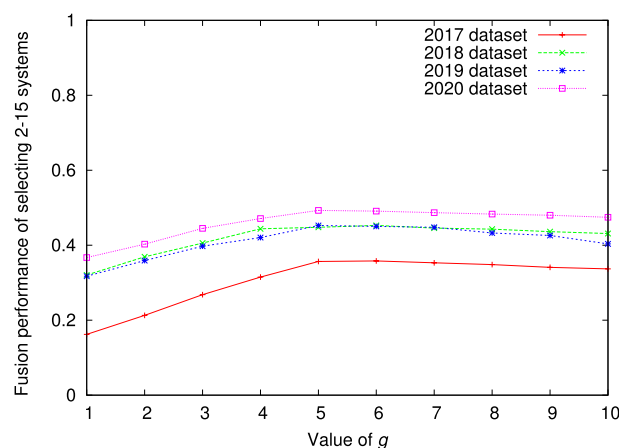


Fig. 2 Fusion performance (MAP) of Algorithm 2 with different g values

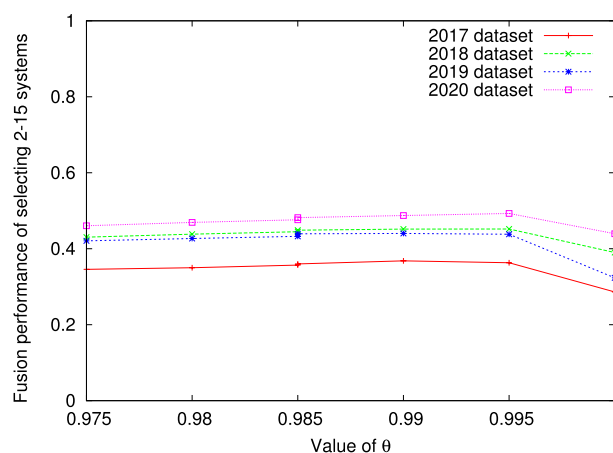


Fig. 3 Fusion performance (MAP) of Algorithm 3 with different θ values

Conclusions

Data fusion has been shown in previous studies to be an effective approach for improving information retrieval performance. However, as the number of component retrieval models increases, the computational cost of fusion also rises. In this paper, we propose three subset selection methods designed to identify an optimal subset from a large pool of retrieval systems or results, enabling efficient yet effective fusion. These methods reduce the computational cost of data fusion while maintaining its performance advantages. Experiments conducted on four TREC biomedical datasets demonstrate that the proposed approaches achieve strong fusion performance and consistently outperform all baseline methods. Notably, even with a small percentage of selected results, the fusion outcomes are significantly better than those of the best individual retrieval system. Although our experiments focus on biomedical datasets, the proposed methodologies are general and can be applied to a wide range of retrieval tasks.

In this work, we mainly considered the fusion method of ComSum(RRF), and CombMNZ(RRF) was briefly reported. It should be interesting to investigate if the proposed methods are applicable to linear combination or other supervised fusion methods. Additionally, as a slightly different goal of maximizing MAP values, we plan to investigate subset selection methods of achieving the best fusion results on P@10. These remain our future work.

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Author contributions

SW and XS and CN and HL designed the work. JS did the experimental work. SW wrote the first draft. SW and CN reviewed the paper.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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